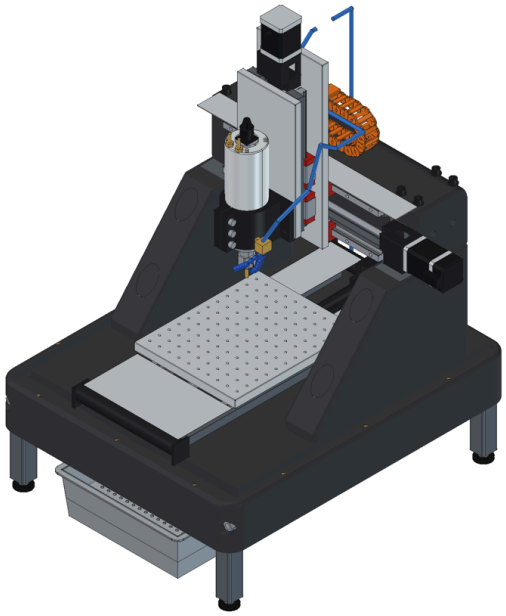
Build guide

DIY CNC epoxy-granite mill · October 2026

How to build the machine in this folder: five epoxy-granite (EG) castings with cast-in steel, HGR15 rails on bedded steel risers, ballscrews on all three axes, a water-cooled spindle, and a 2020/ACM enclosure with flood coolant. Dimensions come from the model (`model/build_cnc.py`, summarised in `docs/5-design/report.txt`); parts and prices are in the sourcing list, and every printed part is in the print guide. Read the whole thing once before buying.



Measure the parts you actually have before cutting or casting anything, especially the rails, blocks, ballscrew kits and spindle clamp. Clone parts can differ from the catalogs by a millimeter or two, and the castings lock in whatever you assume.

Torques below are starting points for these fastener sizes, not manufacturer specs.

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0. Tools, space and safety

- **Space:** a level bench or floor area of about 1.2 x 1.5 m at 20-25 °C for a week. Epoxy cures slowly and badly below 18 °C.
- **Tools:** drill press, chop saw or hacksaw, angle grinder with flap disc, taps (M4, M5, M6), drill/mixer with a paddle (or a small cement mixer), kitchen scale + a 50 kg scale, vibration source (a table vibrator, a concrete poker, or a reciprocating sander with the pad removed pressed against the mold), straightedge (at least 600 mm, ideally ground), feeler gauges, dial indicator with a magnetic base, precision level, torque wrench.

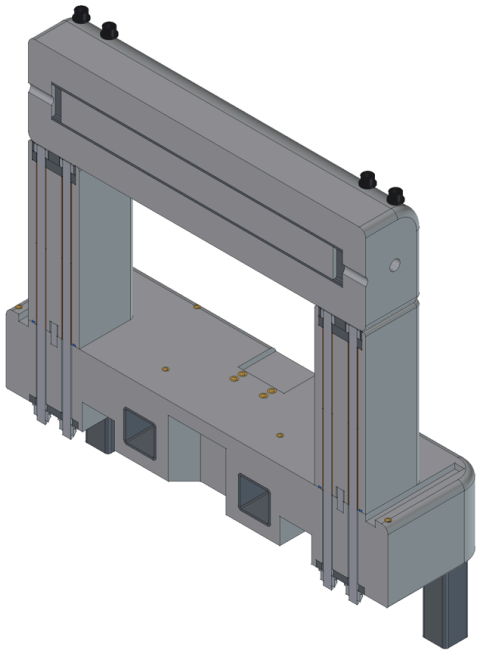
- **Safety:** nitrile gloves and long sleeves for epoxy (skin sensitisation is permanent); an N95 dust mask when pouring the whiting; a P100 respirator when grinding or sanding cured EG (the quartz sand and gravel in it make silica dust; work wet if you can); eye protection when cutting steel. Mix in a ventilated space.
- **3D printing (PETG):** the mold kit (brackets, saddles, holders, bushings), the core posts, formers and fillers, the parts that get cast in (spigot sleeves, tube plugs, drain liner) and the machine's printed parts. All of it is in `print/`, split to fit a Bambu A1 mini, and listed in the print guide: 1.4 kg of PETG (sliced), inside two 1 kg rolls, and about 115 hours of printing. The base pocket cores are cut from one sheet of 2 in XPS foam board instead. Print one bracket and one saddle first and check them on your board; print the sleeves early and test-fit them on your actual tube.
- **Foam cutting:** a fine-tooth saw or a hot-wire cutter, a long knife, a sanding block, packing tape and double-sided carpet tape.

1. Buy and prepare the steel

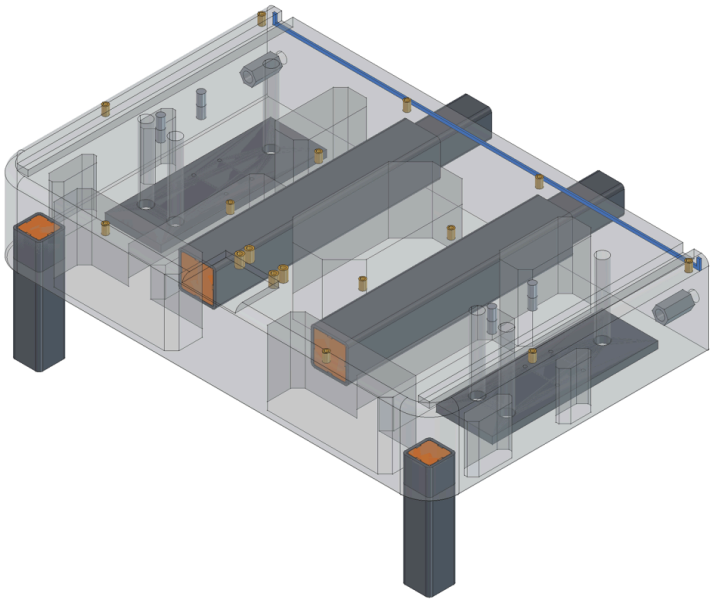
Nothing you order needs cutting: buy the tube at these lengths, and order the risers, the top plates and the aluminum plates cut to size. Only the 1/2-13 threaded rod is cut.

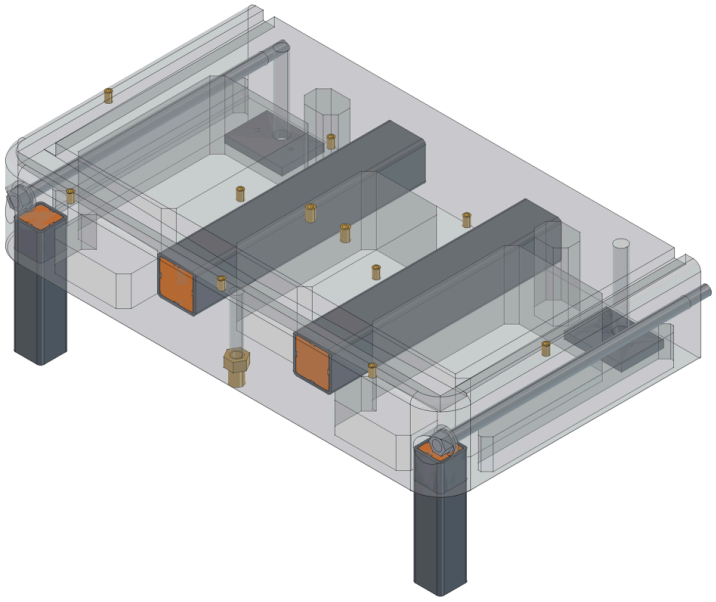
1. **Square tube** (plain steel, not galvanized; 0.120 or 1/8 in wall; remnants are fine):
   - 2 x 2 in: 4 x 12 in (base; they also take the split spigots), 2 x 18 in (beam).
   - 1-1/2 x 1-1/2 in: 6 x 6 in (2 base spigots + 4 legs).
2. **Check the steel when it arrives.** A supplier's saw cut is good to about 1/8 in, and square tube is about 0.5 mm off its nominal size, with a weld seam down the inside. None of that needs fixing, because each piece is set from one end and the other end has room (the "Steel cut to length" table in `docs/5-design/report.txt` lists it):
   - **Length:** anything within 1/8 in of the listed length is fine. Mark one end of each 2 in base tube as its split end, and mark the middle of each spigot.
   - **Ends:** square within a couple of degrees (a 2 in end no more than 1.5 mm out across it). Knock off burrs.
   - **Size:** caliper the outside of each size of tube, and the inside of a 2 in tube away from the seam. The printed parts that fit on or into tube (bushings, leg holders, sleeves, plugs, locators) have 0.5 mm a side and grooves for the seam, so they fit tube that's up to 0.5 mm off. If yours is further off, enter the sizes in `frame_tube` / `post_tube` in `model/build_cnc.py` (with `tube_fit` at 0.25 for a closer fit), re-run it and print those parts from the new files.
   - **Test fit:** a printed sleeve on a spigot, both into a 2 in base tube. If it's tight, file the seam. If it's loose, wrap the sleeve in a turn of tape.
3. **Drill the beam tubes:** each 18 in tube gets 3/4 in holes straight through its top and bottom walls where the inner beam cap screws pass, 4 holes per tube. Measure the tube, mark its middle, and mark the holes 203 mm either side of that mark on the cap-screw line (7 mm off the tube's center line, toward the outside of the beam). Measuring from the middle splits any length error between the two ends, and the 3/4 in holes give the 1/2 in rods that form the bolt holes 3 mm of play a side. Use a drill press, a step drill or a hole saw.
4. **Top plates and anchor bars** (laser cut): upload the DXFs in `docs/1-sourcing/laser/` to a laser-cutting service (SendCutSend, OSH Cut or a local shop). The README there lists them.
   - **Top plates:** two layers per upright, all 1/2 in steel, 68 x 168 mm. The lower layer carries everything: tap its four 5/16 in holes 3/8-16 for the beam screws. The two top tie-rod nuts sit in the upper layer's hex holes, on the lower layer. Keep each pair together; in the mold the greased 3/8 bolts that hang the plate hold the two layers together.
   - **Anchor bars:** 3/8 x 3 in steel, two 75 mm (front half) and two 245 mm (rear half). Tap the small holes M5 (the mold hangs the bars by them). The big holes take the tie rods.

5. **X support pads** (1/4 x 2-1/2 in flat bar, 2 pieces, about 49 and 36 mm long): drill and tap M6 at the BK10 and BF10 bolt positions (46 mm apart across the axis; along it, 13 mm apart for the BK10). These end up flush in the beam face and carry the X ballscrew bearings.
6. **Rail risers** (cold-rolled 1018, 1-1/4 in wide, ordered cut to the rail lengths): Y 3/4 in x 600 mm (2), X 5/8 in x 400 mm (2), Z 5/8 in x 300 mm (2). The rails are used at full length.
    - Clamp each rail on its riser, centerd, and use the rail's own holes as the drill template; drill and tap M4 (60 mm pitch). Mark which rail goes on which riser.
    - Y and X risers also get the hold-down holes, midway between rail holes and under the rail: 5.5 mm through with a 9.5 mm x 4.5 mm deep counterbore from the top. Positions from the riser center: Y at 0, ±120 and ±240 mm, X at ±30 and ±150 mm.
    - Stone or lightly sand off burrs; keep the top faces flat.
7. **Clean all cast-in steel:** wire-brush or flap-disc to dull gray metal, then degrease with acetone just before casting. If it will sit more than a day or two, give it a thin coat of epoxy primer. Paint only the part of each leg that will hang below the base (after casting).
8. **Threaded rod and hardware:** cut the 1/2-13 rods (per upright 2 x 375, 1 x 290 and 1 x 187 mm; 2 x 358 mm for the base draw rods), deburr the ends. A rod that's a few mm off is fine: its top end is set flush and the error ends up in the 6 mm tail under the bottom nut. Have the inserts ready: 18 x M5 (12-16 mm long) and 6 x M6 x 15 mm cast-in inserts, plus 11 x M5 for the enclosure.



The cast-in steel, inserts and printed liners for each base half (castings drawn see-through):





2. Molds and cores

The molds are a kit. Melamine boards are cut to a list and drilled from hole maps, and printed brackets hold them together with M4 bolts. Every printed piece bolts or pins to its board from outside, into a nut built into the piece. Nothing is glued, nothing goes through a cast face, and the molds come apart and go back together the same way.

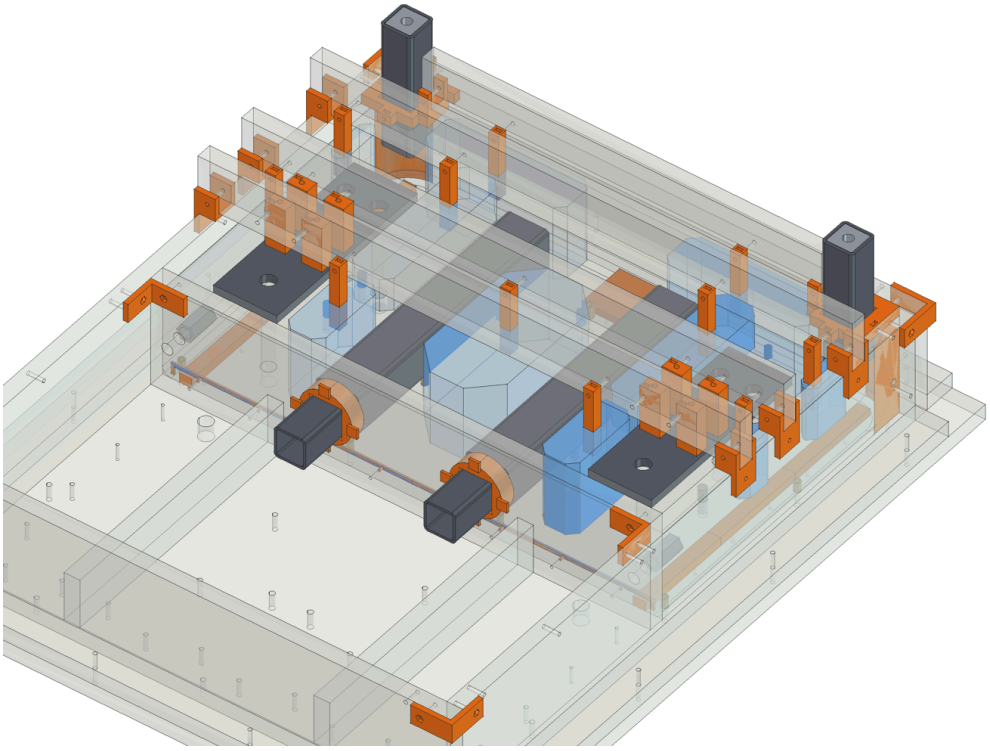
The **mold guide** has the cut list, the hardware, a drawing and a hole table for every board, and the steps (`docs/3-molds/`, one guide per mold); the print guide lists the printed parts.

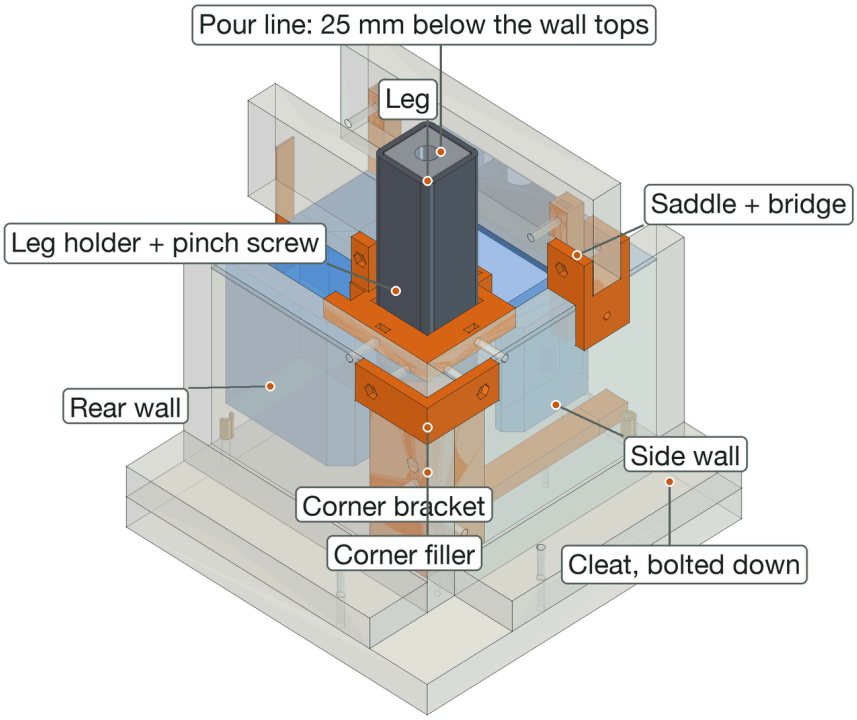
- **Walls stand 25 mm above the pour line.** The brackets, saddles and leg holders bolt through that band. Fill to the line, not to the top.
- **Everything that has to be flat or located goes against the floor**, so each casting is poured with its most important face down.
- **Edges:** the castings' outside edges are rounded 10 mm. Glue 3/8 in quarter-round molding (PVC or primed wood) into the inside corners of the boxes that form outside edges, and run small silicone fillets everywhere else.
- **Release:** wax every surface the mix touches 3-5 times (Partall Paste #2), buff, then spray or wipe PVA. That includes the printed pieces that come out; not the ones that stay in.

Casting	Face on the mold floor	Size (mm)	Mass
Base front	top face (coolant floor)	600 x 345 x 110	30.5 kg
Base rear	top face	600 x 380 x 110	41.9 kg
Upright (x2)	foot (the face that sits on the base)	80 wide, foot 400 long, 246 tall, sloped front	15.7 kg each
Beam	front face (rails)	530 x 140 x 140	26.3 kg

Casting order:

1. **Base rear half**, against a partition at the split, on a floor board big enough for both halves.
2. **Base front half** on the same floor, with the partition out, cast against the rear half's waxed split face. The two top faces come out flush and the split faces match. The front tubes and their sleeves slide straight onto the rear half's spigots, and the draw rods screw into its couplers, so that joint fits by construction. The split's gasket groove is in the rear half; tape over it first.
3. **Uprights**, both from one batch, in two molds.
4. **Beam.**

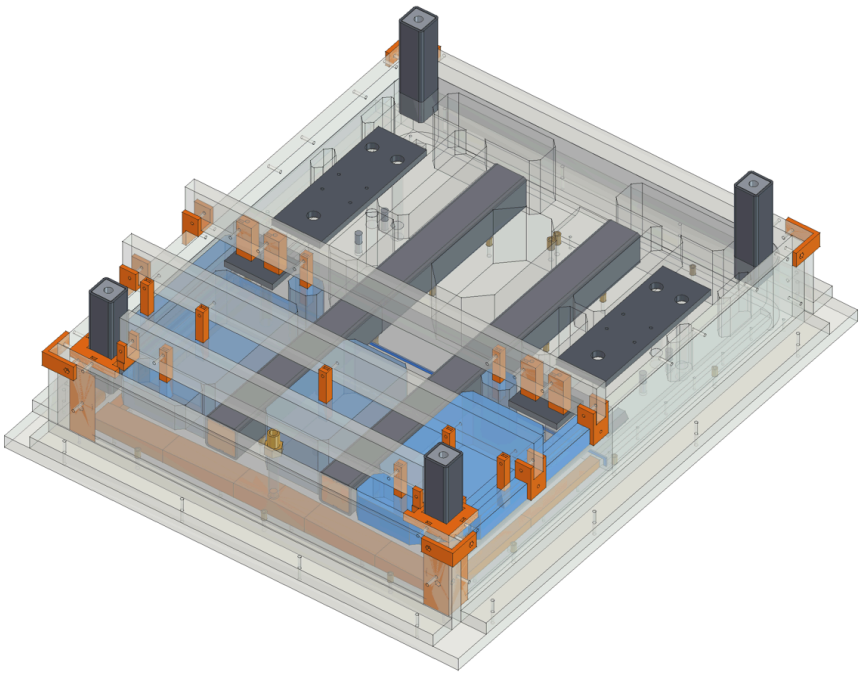


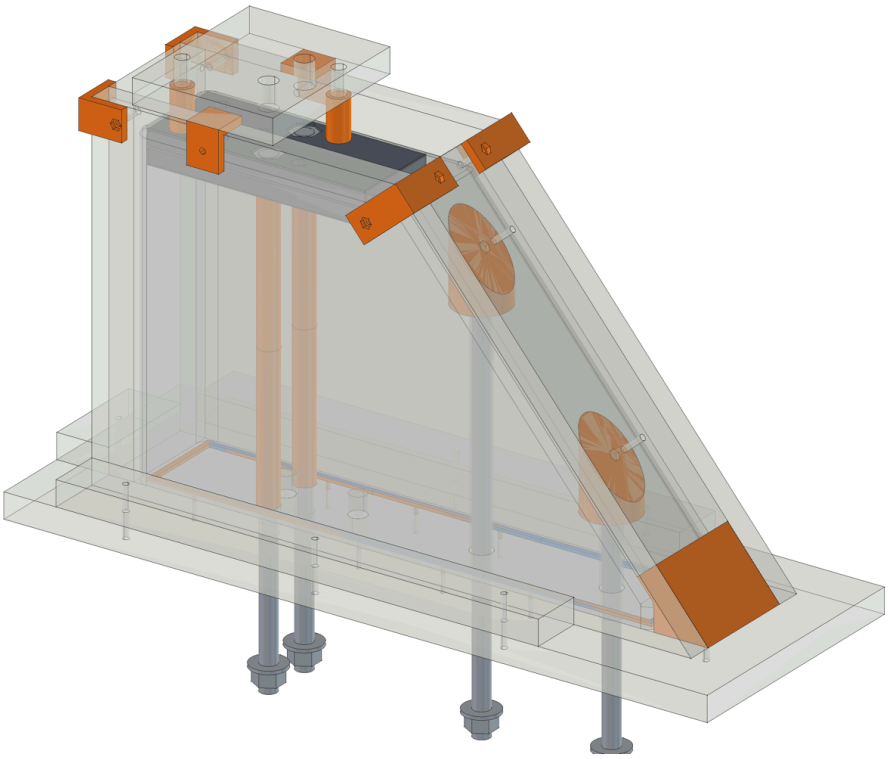


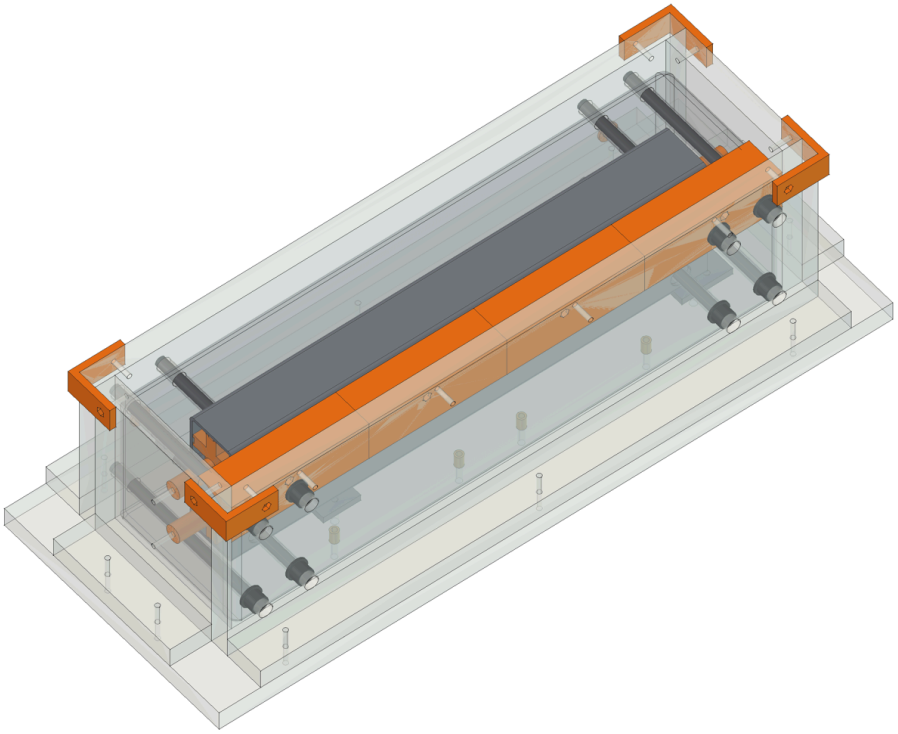
Locating the cast-in parts (the mold guide has the details; grease every screw and rod that comes out):

- **Inserts** (M5 enclosure, M5 riser hold-downs, M6 Y BK/BF): screwed to the floor from underneath, flush with the casting face.

- **Dowels** (rear half): two per upright, pushed halfway into holes in the floor; they stand 15 mm proud and are grouted into oversize holes in the upright's foot at assembly (section 4), so the two molds don't have to agree to a tenth.
- **Tie-rod holes and anchor bars:** greased 5/8 in rods up through the floor form the tie-rod holes right through the base. At the end of the pour the anchor bars go over the rods' tops, flush with the pour line, each screwed (M5) to printed hangers bolted to the bridges.
- **Draw-rod couplers** (rear half): on bolts through the partition.
- **Spigots** (rear half): through printed bushings in the partition, half their length out; the rear tubes, sleeves and end plugs slide onto them.
- **Legs:** through printed holders bolted into the corners above the pour line, each held at its cast-in depth by a pinch screw.
- **Cores:** taped XPS foam blocks on the wet 20 mm skin; a printed post glued into each bolts to the side of a bridge, which sets its height and stops it floating. They can stay in.
- **Uprights:** the top plate (both layers) hangs from the bridge on two greased 3/8 bolts. The two top tie-rod holes are printed sleeves, centered on greased 1/2 in rods pushed up from under the floor through the plate and the bridge; 5/8 in rods form the two slope holes (1.6 mm a side over the tie rods); a greased 1/2-13 bolt forms the two dowel holes (grouted at assembly).
- **Beam:** the tubes hang between printed locators bolted through the end walls (each leaves a 16 mm hole in the beam end). The locators only hold the tube ends; the 1/2 in rods through the tubes' 3/4 in holes set them along the beam, so a tube that's a little long or short doesn't matter. The rods also form the cap-screw holes (1.6 mm a side over the 3/8 in bolts). The support pads and riser inserts screw to the floor; butter each pad with thickened epoxy (epoxy and whiting, peanut-butter thick, about 3 mm) just before the tubes go in, so it fills the 1.5 mm gap to the front tube whatever the tube's size. The dowels sit in the bottom wall on printed stops.







3. Mixing and casting

Use the graded mix, by weight. The aggregate follows a Fuller-type packing curve (each size fills the gaps between the next size up), which is what lets the epoxy stay at 8.5 %:

Part	Size	Source	% of mix	For the whole frame
Epoxy		Raka 127 + 606	8.5 %	10.3 kg
Whiting (calcium carbonate)	325 mesh, under 0.05 mm	pottery supplier	9 %	10.5 kg
Fine sand	0.1-0.6 mm	pool filter sand (#20 silica)	15 %	10 kg (base) + 10 kg of 70 grit
Coarse sand	0.6-3 mm	concrete / all-purpose sand, sieved	27.5 %	19 kg (base) + 20 kg of #16 grit
Pea gravel	3-6 mm	bagged pea gravel, sieved	20 %	23 kg
Gravel	6-10 mm	3/8 in crushed stone (#8) or bigger pea gravel	20 %	23 kg

Two kitchen-type sieves (about 3 and 6 mm mesh, or hardware cloth) split the bagged sand and gravel into these bands. Wash anything dusty or clay-coated, and dry everything completely (spread it out in the sun, or an old oven tray): water stops the epoxy bonding. Crushed stone locks together a little stiffer than rounded gravel; rounded flows and packs a little better; either works.

Beam and uprights: aluminum oxide instead of the sand. Aluminum oxide grit is about five times stiffer than sand or granite. Brown fused #16 (about 1.2 mm) replaces the coarse sand and black 70 grit (about 0.2 mm, Harbor Freight) replaces the fine sand; together they make these castings roughly 35 % stiffer, about 17 % on the whole frame. The grit is heavier, so swap it by volume: **1.5 kg of #16 for every 1 kg of coarse sand, and 1.4 kg of 70 grit for every 1 kg of fine sand**, everything else (epoxy included) the same. A 50 lb bag of #16 and a 25 lb bag of 70 grit cover the beam, both uprights and a pair of test bars. The base keeps the sand. Don't over-vibrate these: the heavy grit can settle. Use an old paddle; the grit wears it.

8.5 % only works if the aggregate packs densely, so prove the mix before casting anything:

1. **Jar test.** Fill a 1 L jar with the dry aggregate blend (no whiting), shaking and tapping it down, then pour in water from a measuring cup until it reaches the top. The water is the void space. Nudge the coarse sand and fine sand up or down a few percent until it's as small as you can get it (below about 25 % of the jar is good).
2. **Test bars.** Print or build a 40 x 40 x 500 mm bar mold (waxed like the others) and cast two bars at 8.5 % and two at 9.5 %, about 2 kg each, vibrated the same way as the castings, plus two at 8.5 % with the alumina grits for the beam and uprights.
3. **Check them after 7 days.** Cut one bar of each pair open: there should be no voids larger than a grain of sand, and every stone should be wetted. Then tap-test the other: weigh it, rest it on two foam strips 112 mm in from each end, tap the middle with a small hammer and read the lowest strong peak on a phone spectrum app (phyphox, Spectroid). Its stiffness is about **$E\ (\mathrm{GPa}) = 4.8\mathrm{e}{-5} \times \mathrm{mass\ (kg)} \times \mathrm{frequency\ (Hz)}^2$**. The frame model assumes 30 GPa, about 570 Hz for a 1.9 kg bar; 25 GPa would cost about 13 % of the frame stiffness.
4. **Pick the mix.** Use 8.5 % if its bar is void-free and at least as stiff as the 9.5 % one; otherwise use 9.5 % and order the extra 1.5 qt epoxy kit. The alumina bar should come out about 35 % stiffer than the plain 8.5 % one (it's heavier, so use the formula, not the pitch); if it's well short, or it has voids, cast the beam and uprights with sand too.

Casting	Mix	Epoxy (8.5 %)
Base front	30.5 kg	2.6 kg
Base rear	41.9 kg	3.6 kg

Casting	Mix	Epoxy (8.5 %)
Uprights (alumina)	2 x 15.7 kg	2 x 1.2 kg
Beam (alumina)	26.3 kg	1.9 kg

Add 5-10 % for waste: about 11 kg, plus about 1 kg for the test bars, which fits the 3 gal kit (about 12.5 kg). Mix only what you can place; if you're running short, the 1.5 qt kit covers the gel coat.

1. **Dry and clean** all aggregate. Weigh everything out per batch before opening the epoxy. Keep batches to what you can place within about half the epoxy's pot life (10-15 kg each).
2. **Mix order:** epoxy resin + hardener (exact ratio, by weight), then the whiting until smooth, then the sands, then the gravel. Mix until every stone is wet and the color is even.
3. **Place in layers** of 30-40 mm. Vibrate each layer until the surface goes glossy and stops releasing bubbles, then add the next. Work the mix around the steel, inserts and cores with a stick. Vibrate the full mold for 20-30 minutes at the end.
4. **Level the open face** at the pour line, 25 mm below the wall tops: strike it off with a straight board in the two printed screed shoes (mold guide, "Pour"). Cover against dust, and leave it at 20-25 °C. Demold after the time your epoxy needs (usually 24-48 h), but don't load or machine anything for 7 days.
5. Pull all the greased screws and rods, the formers, fillers and plugs. Printed cores stay in; foam cores can be dug out or dissolved with a little acetone. Unbolt the mold kit; it's reused.
6. **Seal** the base top (the coolant floor) and the gutters with a thin epoxy gel coat. Paint the exposed leg ends.

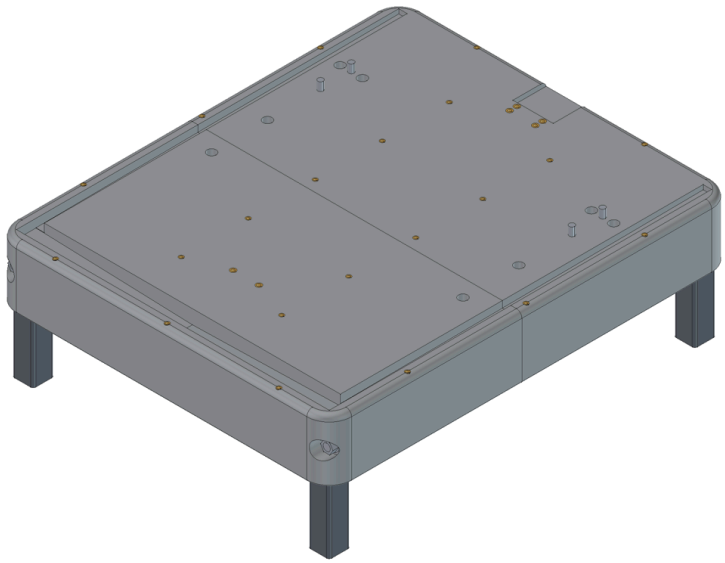
Cast the rear base half first and the front half against it (above). Cast the two uprights together from one batch, before the beam, so you can check the top-plate spacing against the beam mold.

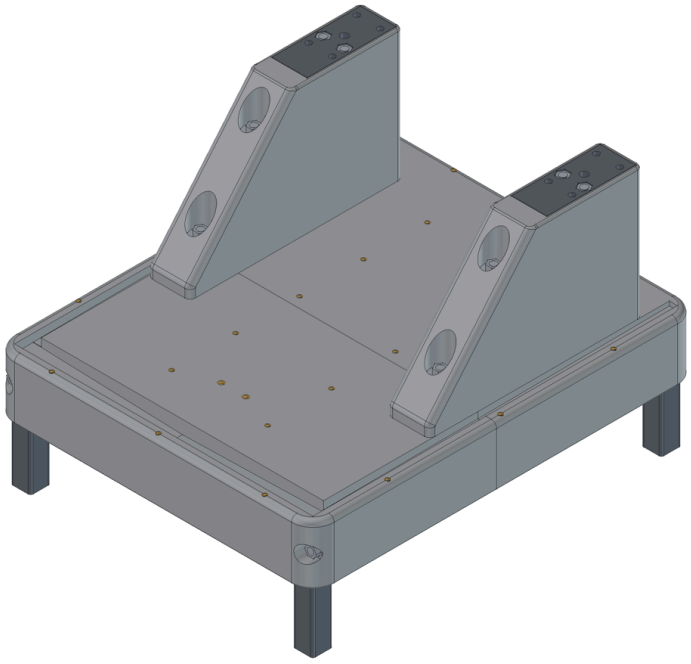
4. Assemble the frame

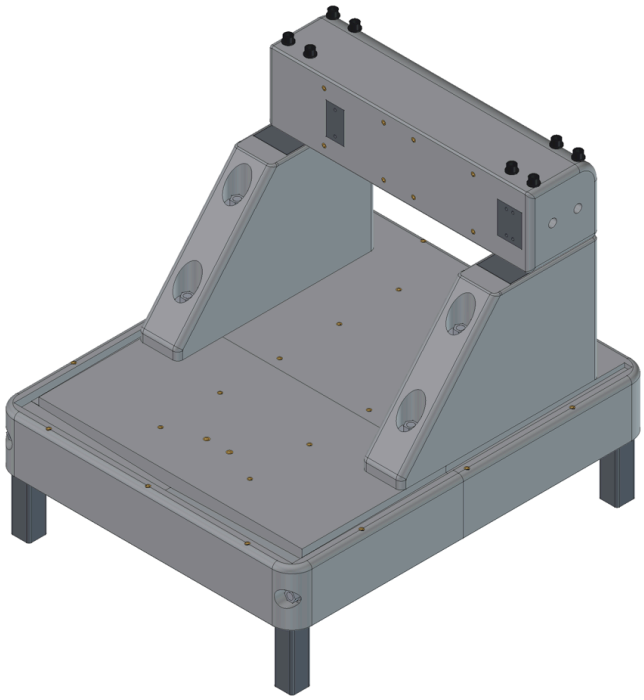
1. **Feet:** set the rear base half on the bench on its legs. Bond a 1/2-13 nut into a printed plug in each leg end (or use the threaded inserts), screw in the leveling feet, and roughly level it.
2. **Join the base halves:** glue the O-ring cord into the rear half's split groove (glue the cord ends). Push the front half on (its tubes and sleeves over the spigots) and tighten the two 1/2-13 draw rods from the front pockets (snug, then about 40 N·m). Level the whole base with the feet.
3. **Grout the dowels.** The two dowel holes in each upright's foot are 1/2 in, over the 3/8 in dowels, so the fit comes from epoxy, not from the two molds agreeing. Wax the base dowels (three coats, buffed). Mix a little of the leftover epoxy with whiting to a thick cream and half-fill each dowel hole. Glue the foot O-ring into each upright's groove, lower the upright on (dowels into their holes, foot on the base) and put the tie rods in (next step) snug, wipe off what squeezes out, and leave it 24 h. After that the upright lifts off and drops back on in the same place.
4. **Tie rods.** Each upright is held by four 1/2-13 rods that run from its top plate or its sloped front right through the base to the anchor bar underneath, so it's steel from the beam screws to the nuts under the base, and every nut can be reached and retightened.
    - **Top two:** drop a nut into each hex hole in the top plate. Push the 375 mm rod up from under the base, through the anchor bar, the base and the upright, and screw it into that nut until its end is flush with the top of the plate (the hex hole stops the nut turning).
    - **Slope two:** fender washer and nut in each slope pocket; push the rod up from below and screw it into the nut until its end is flush with the nut's top.

- **Underneath:** an SAE washer and a nut on each rod against the anchor bar. Snug all four, then tighten to about 40-50 N·m, front to back. Push the printed caps into the two sloped pockets, flush with the face (a dab of silicone holds them).
- Check them again after a week and after the first month; later, once a year.

5. **Beam:** the top plates' dowel holes are 14 mm, oversize the same way. Wax the beam's dowels, put grout in the holes, set the beam on and fit the eight 3/8-16 x 6-1/2 in bolts snug. Check the beam face is square to the base top with a square and a dial indicator (shim under one top plate if needed) and leave it 24 h before the final torque (about 40 N·m).



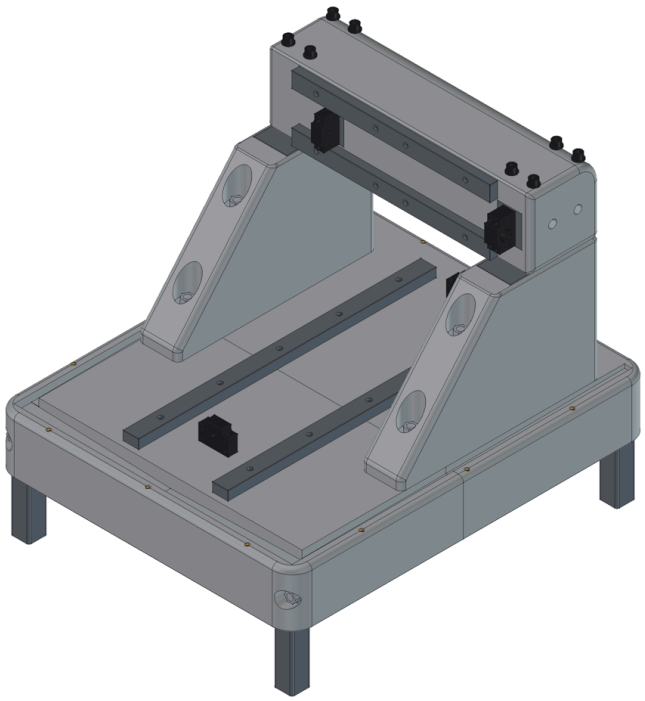




5. Bed the rail risers

The risers are bonded to the castings with a thin epoxy grout and held down by screws into the cast-in inserts. Nothing is screwed into the cast-in tubes; the tubes stiffen the casting and form the joints.

1. Abrade and degrease the casting under each riser and the riser's underside.
2. Set the riser on small shim stacks (feeler stock) at each hold-down. Fit the M5 low-head screws (M5 x 25 for Y, M5 x 20 for X) just snug.
3. **Y risers (base):** make each one straight along its length and both coplanar with each other, within 0.02 mm over the length if you can. Check with the straightedge and feelers, or a dial indicator on a bridge spanning both. Set them parallel, 180 mm apart on center (rail spacing), using the rails as a gauge.
4. **X risers (beam face):** same, 100 mm apart. Make them parallel to the base top, checked with the indicator on the Y-axis table later, or now with a height gauge from the base top.
5. Dam the ends and sides with tape or plasticine, and flow low-viscosity epoxy thickened with a little whiting underneath from one side until it comes out the other.
6. After 48 h, torque the hold-downs to about 4 N·m (brass inserts) and recheck flatness.



6. Axes

Y (table).

1. Bolt the Y rails (full 600 mm, centerd 20 mm behind the base split) to their risers (M4 x 16, about 4 N·m). Tighten the first rail against a straightedge, then set the second parallel with a dial indicator on a block of the first (within 0.01-0.02 mm over the travel).
2. Bolt the BK12 (fixed bearing, rear) and BF12 (front) to their M6 inserts (M6 x 40, about 6-8 N·m) with the screw installed. Align the screw parallel to the rails in both directions with a dial indicator on the nut. Shim under a support if needed; the screw should spin freely by hand along the whole length.
3. Mount the DSG16H nut housing under the table (shim to height), bolt the table to the four blocks and the housing, and check for binding at both ends of travel. Fit the motor bracket, the 8 x 10 coupler and the closed-loop motor.

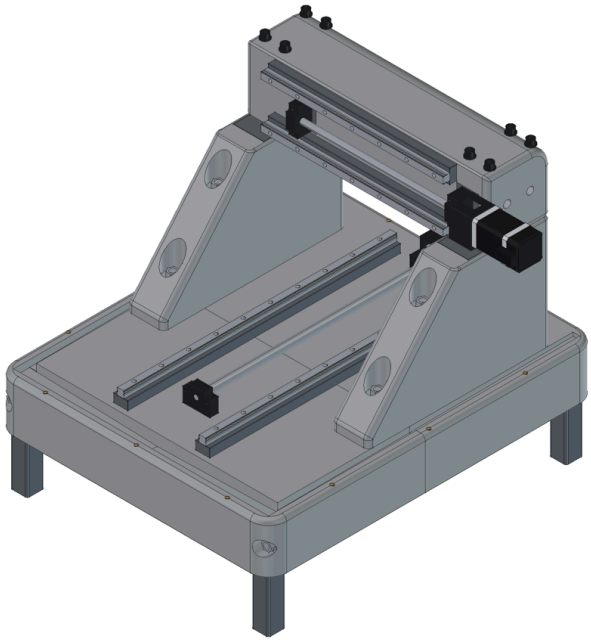
X (beam).

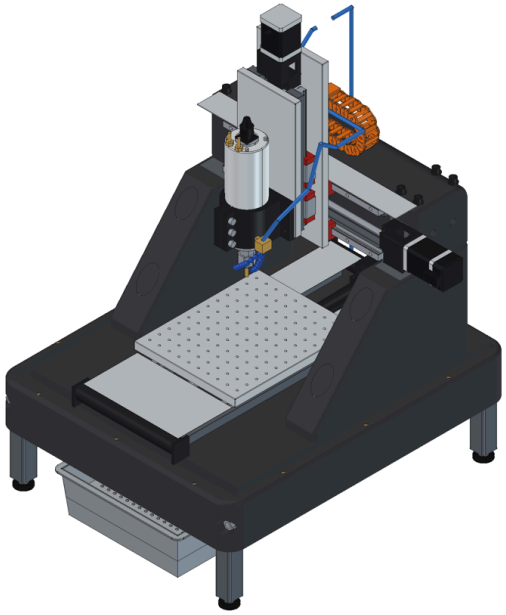
1. Rails onto the X risers as for Y (100 mm spacing).
2. BK10 and BF10 onto the beam pads: M6 x 35 with **two DIN 433 small washers under each head** (they fit the support's counterbore), so the screws stop short of the tube behind the pad. Align and check as for Y.
3. Bolt the X carriage plate to the four blocks and the nut housing; motor bracket, 8 x 8 coupler, closed-loop motor.

Z.

1. Z risers onto the Z plate: tap the risers M5 and screw them on from the back of the plate (no inserts needed in aluminum). Rails on the risers, and the four Z blocks on the front of the X carriage.
2. BK10 / BF10 and the Z screw on the carriage, nut housing on the Z plate, open-loop motor on top.
3. Spindle clamp on the Z plate, spindle in the clamp. Tram it later (section 9).

Lubricate every block with the grease gun before moving anything far.





7. Electronics and spindle

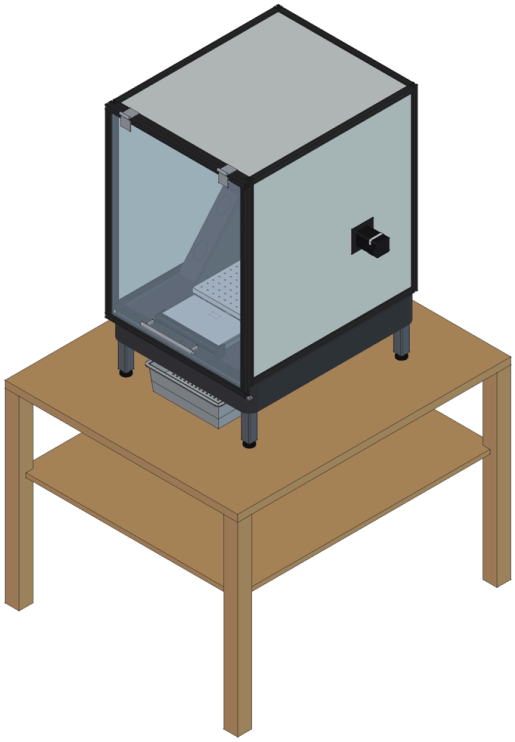
- **Controller:** FluidNC board with external drivers. CL57T drivers for X and Y (closed loop), DM542T for Z, 48 V supply trimmed to about 46 V (the CL57T maxes out at 48 V), 24 V logic.
- **Steps:** with drivers at 1600 pulses/rev: Y (SFU1605, 5 mm lead) 320 steps/mm; X and Z (SFU1204, 4 mm lead) 400 steps/mm.
- **Spindle:** VFD for 120 V now (limits it to about 1.5 kW), the kit's 220 V VFD later on 240 V. Use a dedicated 20 A circuit and shielded spindle cable, earthed at the VFD end. Set the VFD for the spindle's nameplate: 400 Hz, 220 V, 24,000 rpm.
- **Spindle cooling:** water pump in a 10-20 L bucket of distilled water with 10-20 % RV antifreeze, switched by a relay so it runs whenever the spindle runs. Never run the spindle dry.
- **Safety:** wire the E-stop to cut driver enable and the VFD run signal in hardware. Fit the homing switches, then the tool setter.

8. Enclosure and coolant (part 3 of the sourcing list)

The machine runs without these: until then, brush or drip cutting fluid onto the cut, keep the way covers and screw cover on, and let the gutters and drain carry the fluid to the pan under the base. Cast the enclosure inserts into the base now so the frame bolts straight on later.

1. Bolt the 2020 frame to the base top with angle brackets into the cast-in M5 inserts (M5 x 35). Cut list in `docs/5-design/report.txt`.
2. ACM side, back and roof panels; polycarbonate door on hinges and gas struts, magnetic catch. Brush seals where the X and Y motors pass through the panels.

3. Way covers: the stainless strips on their spring rollers in front of and behind the table, the spiral cover on the Y screw, the drip shield over the X rails.
4. Coolant: slide the sump (steam-table pan with the chip basket, magnet, baffle, screen and 24 V pump) in under the base below the drain nipple. Run the return hose to the bulkhead, up the cable chain, to the manifold and nozzles on the spindle clamp. Use water-soluble coolant.



9. Commissioning

1. **Run each axis by hand,** then with the motors at low speed, end to end. Listen for tight spots.
2. **Homing and soft limits** in FluidNC; check the travel (180 / 300 / 140 mm).
3. **Steps per mm:** move 100 mm and measure with the dial indicator or a caliper; correct.
4. **Squareness:** sweep an indicator in the spindle along a straightedge or the table edge. X to Y is set by the beam position (step 4 of the frame); Z to the table is set by shimming the spindle clamp (tram). Aim for under 0.02 mm over 100 mm.
5. **Face the table:** take a light fly-cut over the aluminum table so it is flat and parallel to X/Y.
6. **First cuts:** aluminum first, then steel at light depths of cut with flood coolant. Check the spindle temperature and the coolant flow, and re-torque the frame bolts and the riser hold-downs after the first few hours of running.